

AI Solution Architect - Day 1

Topic: How Computers Really Work

Key Concepts:

- A computer is a programmable machine that executes instructions.
- Internally it is built from billions of transistors.
- A transistor is a tiny electronic switch.
- Each transistor has two states: 0 or 1 (Binary).
- One binary digit is called a Bit.
- Eight bits make one Byte.
- One Byte has $2^8 = 256$ possible combinations and can represent values 0-255.
- Computers do not store letters, images or videos directly; they store only binary data (0s and 1s).

Revision

1. Computer = Executes instructions using billions of transistors.
2. Transistor = Electronic switch.
3. Bit = Smallest unit of information (0 or 1).
4. Byte = 8 bits.
5. One byte can represent 256 values because $2^8 = 256$.

Question & Answer Review

Student Answers:

- Computer is a huge collection of billions of electrical switches.
- Transistor is an electrical switch.
- Bit has two states (0 and 1).
- Byte is 8 bits ranging from 0-255.
- $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 256$.

Instructor Feedback:

- Excellent understanding (Overall Score: 9.6/10).
- More precise definition: A computer is a programmable machine that executes instructions, built using billions of transistors.

ASCII Discussion

Question: Why is 'A' represented by 65?

Answer:

- ASCII (American Standard Code for Information Interchange) is a globally accepted character encoding standard.
- The values were chosen by agreement (a standard), not by nature.
- Standards enable different computer systems to communicate consistently.

Next Lesson Preview:

- Binary numbers
- ASCII vs Unicode
- UTF-8 and UTF-16
- Why emojis require multiple bytes
- Foundation for AI tokens and Large Language Models.